IMO 2026 — Problem 2

Solution by DeepSeek V4 Pro

2026-07-21

Source: `results/deepseek-v4-pro/problem-02/current.md` — autonomous run, no human intervention; solution as written by the model.

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC, respectively. Let points K and L be chosen inside triangles BMC and BNC, respectively, such that K lies inside the angle LBA, L lies inside the angle ACK, and $\angle KBA = \angle ACL$, $\angle LBK = \angle LNC$, $\angle LCK = \angle BMK$. Let O be the circumcentre of triangle AKL. Prove that $OM = ON$.

Solution document

Status

partial

Problem

Let ABC be a triangle and let points M and N be the midpoints of sides AB and AC, respectively. Let points K and L be chosen inside triangles BMC and BNC, respectively, such that K lies inside the angle $\angle LBA$, L lies inside the angle $\angle ACK$, and

$$\angle KBA = \angle ACL, \qquad \angle LBK = \angle LNC, \qquad \angle LCK = \angle BMK.$$

Let O be the circumcentre of triangle AKL. Prove that $OM = ON$.

Approaches tried

- Numeric search + gradient descent: solutions exist and satisfy OM=ON to within numerical error (~1e-7). Earlier wrong circumcenter function gave false non-zero; corrected function confirms OM=ON.
- Sympy polynomial algebra: derived the condition OM=ON is equivalent to numerator P = 0, where P is a polynomial of degree 3 in coordinates (x1,y1,x2,y2,a1,a2). Trying to prove P is in the ideal generated by the three angle conditions.

- Special case (right isosceles A=(0,0), B=(2,0), C=(0,2)): solved exactly. Found two solution branches: one has x1=y2 (which forces OM=ON), the other also forces OM=ON after substituting interior conditions.
- Attempted to find geometric meaning: spiral similarities centered at O or at A don't match. The three conditions might correspond to an isogonal conjugate or a special point.

Current best

Fixed coordinate setup: M=0, N=1, B = -A, C = 2 - A. Then: - OM = ON $2 \cdot \mathrm{Re}(O) = 1$ P = 0 where P = numerator of (2X-1). - P = a1^2*y1 - a1^2*y2 - a1*y1 + a1*y2 + a2^2*y1 - a2^2*y2 - a2*x1^2 + a2*x1 + a2*x2^2 - a2*x2 - a2*y1^2 + a2*y2^2 + x1^2*y2 - x1*y2 - x2^2*y1 + x2*y1 + y1^2*y2 - y1*y2^2

The three angle conditions are (with U=K+A, V=L+A): I: Im(A(A-1) / (U(V-2))) = 0 II: Im(U(L-1) / (V(1-A))) = 0 III: Im(A(U-2) / (K(V-2))) = 0

These give three polynomial equations R1=R2=R3=0.

Need to show P Ideal(R1,R2,R3) or at least that P=0 when the interior conditions hold.

Full proof

(TODO)